3.5 Higher-Order Derivatives

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

What does it mean to take the derivative of a derivative?

Targets

- I can name f?, f? and f? and say what each one tells me.
- I can find a second or higher derivative with the usual rules.
- I can find d^2y/dx^2 at a point on an implicit curve.

Concepts

- **Notation** (f???(x) = d?y/dx?): f?, f?, f? are the first, second and third derivatives. d^2y/dx^2 is the same as f? (or y?).
- **The second derivative** (f? = (f?)?): f? is the derivative of f?. It is the slope of f?: how the slope of f is changing.
- **Reading the sign** (f?(x) = slope of f?): f? >0 means f? is increasing. f? <0 means f? is decreasing.
- **Patterns save work** (d?/dx? e^(kx) = k? e^(kx)): A polynomial loses one degree each time. sin and cos repeat every 4 derivatives. Each derivative of e^(kx) multiplies by k.
- **Implicit curves** (y? = d/dx (dy/dx)): Differentiate dy/dx again. Each y term brings a new dy/dx. Then put in the dy/dx you already found.

What the class predicted, and what is true

- The derivative of x^2 is $2x$. The derivative of $2x$ is... **2.** The line $2x$ has slope 2, so differentiating again gives 2.
- Keep differentiating $\sin x$: $\sin x$, $\cos x$, $-\sin x$, $-\cos x$, and then... **sin x.** The next derivative of $-\cos x$ is $\sin x$. The pattern repeats every 4 derivatives.

Huddle notes (answers filled in)

Definitions

- **Higher derivative:** Differentiate **again**. The second derivative is f?(x) or d^2y/dx^2. The third is f?(x).
- **What f? means:** f? is the **slope** of f?. It tells how the slope of f is **changing**.
- **Reading the sign:** If f? >0, the slope of f is **increasing**. If f? <0, it is **decreasing**.
- **Patterns:** A polynomial loses **one** degree each time. sin x repeats every **4** derivatives.

Worked example

Let f(x) = x? – $3x^2$ + 2x. Find f? and f?.

- First derivative: f?(x) = $4x^3$ – 6x + 2
- Second derivative: f?(x) = **12**x^2 – 6
- Third derivative: f?(x) = **24**x
- The fourth derivative is the constant **24**

Label the three graphs

1. The cubic is the graph of **f**.
2. The parabola is the graph of **f?**.
3. The straight line is the graph of **f?**.

One worked sample of each skill

Second derivative

Level 1. Find the second derivative.
Answer: $24x + 8$ $f'(x) = 12x^2 + 8x + 3$, so $f''(x) = 24x + 8$.

Level 2. Find the second derivative.
Answer: $4e^{2x}$ $f'(x) = 2e^{2x}$. Differentiating that again gives the answer.

Level 3. Find the second derivative.
Answer: $\left(2 + 4x^2\right) e^{x^2}$ $f'(x) = 2x\, e^{x^2}$. Differentiating that again gives the answer.

Level 4. Find the second derivative.
Answer: $\left(x^2 + 4x + 2\right) e^{x}$ $f'(x) = \left(x^2 + 2x\right) e^{x}$. Differentiating that again gives the answer.

Higher derivative at a point

Level 1. $f(x) = e^{2x}$. Find $f^{(4)}(0)$.
Answer: 16 Each derivative multiplies by 2. After 4 derivatives: $2^4 e^{2x}$. At $x = 0$ that is 16.

Level 2. $f(x) = \sin x$. Find $f^{(99)}(0)$.
Answer: -1 The pattern repeats every 4. 99 leaves remainder 3, so $f^{(99)}$ is the same as $f^{(3)}$. At $x = 0$ the value is -1.

Level 3. $f(x) = \cos(2x)$. Find $f^{(6)}(0)$.
Answer: -64 Each derivative multiplies by 2, so the size is $2^6 = 64$. The sign follows the cos pattern: $-$. The answer is -64.

Level 4. $f(x) = \cos(3x)$. Find $f^{(4)}(0)$.
Answer: 81 Each derivative multiplies by 3, so the size is $3^4 = 81$. The sign follows the cos pattern: $+$. The answer is 81.

Read the second derivative

Level 1. $f''(x) < 0$ on an interval means the slope of f is...
(A) increasing **(B) decreasing** (C) constant (D) undefined
Answer: B f'' is the slope of f'. Positive: f' goes up. Negative: f' goes down. Zero: f' stays level.

Level 2. $f'(x) = x^2 - 4x - 4$. At $x = 4$, is f'' positive, negative or zero?
(A) positive (B) negative (C) zero (D) cannot tell
Answer: A $f''(x) = 2x - 4$. At $x = 4$ it equals 4, which is positive.

Level 3. The table gives two values of f'. Estimate $f''(4.5)$ with the slope between them.

Answer: -5 Rise over run: $\dfrac{-34-(-9)}{7-2} = \dfrac{-25}{5} = -5$.

Level 4. The table gives two values of f'. Estimate $f''(2.5)$ with the slope between them.

Answer: 6 Rise over run: $\dfrac{36-(6)}{5-0} = \dfrac{30}{5} = 6$.

Second derivative of a curve

Level 1. The curve $x^2 + y^2 = 16$ passes through $(0, 4)$. Find $\dfrac{d^2y}{dx^2}$ at that point.

Answer: -0.25 $y' = -\dfrac{x}{y} = 0$ at the point. Then $y'' = -\dfrac{y - xy'}{y^2} = -\dfrac{y}{y^2} = -\dfrac{1}{y}$ there, which is -0.25.

Level 2. The curve $xy = 4$ passes through $(1, 4)$. Find $\dfrac{d^2y}{dx^2}$ at that point.

Answer: 8 $y' = -\dfrac{y}{x} = -4$ at the point. Differentiating again: $y'' = -\dfrac{y'x - y}{x^2} = 8$.

Level 3. The curve $x^3 + y^3 = 9$ passes through $(2, 1)$. Find $\dfrac{d^2y}{dx^2}$ at that point.

Answer: -36 Find y' first, differentiate it again (each y brings a new y'), then put in y' and the point. The answer is -36.

Level 4. The curve $x^3 + y^3 = 9$ passes through $(2, 1)$. Find $\dfrac{d^2y}{dx^2}$ at that point.

Answer: -36 Find y' first, differentiate it again (each y brings a new y'), then put in y' and the point. The answer is -36.